Supplementary material for: A physics-informed digital twin for tube-life-aware operation of industrial steam methane reformers under uncertainty

Izbassar Orynbassar[a,*], Madina Sissenbay[b]

[a]School of Information Studies, Syracuse University, Syracuse, NY, USA
[b]Institute of Petrochemical Engineering and Ecology named after N.K. Nadirov, Safi Utebayev Atyrau University of Oil and Gas, Atyrau, Kazakhstan

S1. Model implementation detail

Effectiveness factors. Intraparticle diffusion limitation is severe in reformer catalysts. Rather than solving the pellet problem, the constant effectiveness factors adopted by Latham et al. [3] following Wesenberg and Svendsen [4] are used: $\eta_j = 0.1$ along the tube and 0.05 in the first tube section (10 % of the length here), where the feed contains little hydrogen and the rate expressions are stiff.

Higher alkanes at the inlet. Higher alkanes in the feed (C_2–C_5) are converted at the inlet by the overall rule used by Latham [3],

$$\mathrm{C_kH_{2k+2}} + \tfrac{k-1}{3}\,\mathrm{H_2O} \rightarrow \tfrac{k-1}{3}\,\mathrm{CO} + \tfrac{2k+1}{3}\,\mathrm{CH_4}, \tag{1}$$

whose enthalpy is small because the endothermic reforming and the exothermic hydrogenolysis nearly cancel.

Equilibrium constants and the hydrogen floor. The equilibrium constants computed from Gibbs energies of formation agree with the common empirical expressions within 7 % between 800 and 1100 K. A floor of 10^{-6} bar on $p_{\mathrm{H_2}}$ avoids the singularity of the rate expressions at a hydrogen-free inlet.

Heat-release profile and furnace-side convection. The heat-release density $r(z)$ of the long-furnace model is a downward-opening parabola that vanishes at $z = L_qL$, releases the fraction α_{top} of the heat in the top fifteenth of the furnace height and integrates to $(1 - f_{\mathrm{loss}})\,Q_{\mathrm{comb}}$, with $f_{\mathrm{loss}} = 0.02$ for casing losses; this is the continuous form of the sectional heat-release profile of Latham [3]. The furnace-side convective coefficient h_c is taken from the Dittus–Boelter correlation on the flue-gas flow area, scaled by the factor $f_{\mathrm{ctube}} = 0.38$ fitted by Latham et al. [3]; at furnace-gas temperatures the convective term is a minor correction to the radiative flux. The combustion heat Q_{comb} and the flue-gas flow and composition follow from the lower heating value and complete combustion of the fuel gas and pressure-swing-adsorption off-gas with the specified excess air.

Coupled solution. The outer-wall temperature at each height is found by a bracketed root search on the flux balance, evaluated inside the right-hand side of the ODE system, so that the furnace and tube equations are integrated together in a single downward sweep.

Thermal stress across the tube wall. The elastic thermal stress from the radial temperature gradient, $E\alpha\Delta T_{\mathrm{wall}}/[2(1-\nu)]$, is about 45 MPa for the 20–30 K wall drop of the calibrated tube ($E \approx 150$ GPa, $\alpha \approx 18 \times 10^{-6}$ K^{-1} at 900 °C, $\nu = 0.3$), several times the 12 MPa hoop stress.

Thermal time constants. The thermal time constant of the tube wall, $\rho c\, t^2/\lambda$, is of the order of one minute; that of the refractory lining is of the order of a day.

*Corresponding author.
Email address: iorynbas@syr.edu (Izbassar Orynbassar)

Calibration procedure. The nine residuals per plant case are the outlet temperature, the outlet pressure, the outlet wet composition, the flue-gas outlet temperature and the two wall temperatures. The trust-region-reflective algorithm with bounds is started from a Latin-hypercube set of initial points to guard against local minima, and the parameter covariance is obtained from the Jacobian at the optimum with t-based 95 % intervals.

S2. Operating space and campaign design

Table S1: Operating space sampled in the design, sensitivity, surrogate and optimisation campaigns. The base column is the calibrated Plant A state. Ranges are motivated by industrial practice and by the operating points of the sources used for verification and calibration.

Variable	Unit	Min	Max	Base	Basis
steam-to-carbon ratio	–	2.5	4.0	2.89	carbon-formation limit on Ni catalyst below about 2.5; upper value typical of hydrogen plants; 2.7–3.0 in the reference plant [3], 3.36 in [2]
feed per tube	fraction of base	0.60	1.10	1.00	industrial turndown of about 50–60 %; 110 % short-term overload
specific firing Q_{comb}/feed	fraction of base	0.85	1.15	1.00	burner control range around the plant value
inlet temperature	K	800	900	884.5	884.5 K in the reference plant, 793 K in [2]; cracking in the preheat coil above about 920 K
inlet pressure	bar	25	33	30.1	industrial range 20–40 bar; 28–30 bar in the reference plant, 29 bar in [2]
catalyst activity	–	0.10	0.30	0.20	0.20 fitted to the reference plant for a mid-campaign catalyst [3]; fresh higher, end-of-campaign lower
excess air	%	5	25	21.6	industrial reformer furnaces 5–20 %; upper value extended so that the calibrated base lies inside the sampled space

The campaigns evaluated on this space are: an open-loop Latin-hypercube design of 2000 points (surrogate training); a Saltelli design of 4096 points for the Sobol indices and as an independent surrogate test set; a closed-loop 15×15 grid in steam-to-carbon ratio and load at constant outlet temperature (Fig. 5 of the main text); twenty scenario-years of 8760 hourly states; an NSGA-II run of 200 individuals over 300 generations; and a scrambled Sobol Monte Carlo design of 2048 samples propagated through the physics model at eight operating points, with 1024 further samples per uncertainty group for the variance decomposition. All designs use seed 0. The complete pipeline runs in 35 minutes on a ten-core laptop.

Table S2: Definitions of the six annual scenarios of Section 2.7 of the main text. Each is an 8760-hour hourly history evaluated under one of the two control modes.

Scenario	Control mode	Definition
S1 steady base	either	constant operation at the calibrated Plant A state
S2 daily load-following	either	full load 06–22 h, 70 % at night, ramps of 0.10 per hour
S3 renewable-following	either	bounded Ornstein–Uhlenbeck load process between 0.60 and 1.10 with mean 0.85
S4 over-firing events	either	24 events per year of $+10$ % firing for 2 h; six events of $+25$ % for 1 h; a flame-impingement proxy adding 40 K to the hot spot for 24 h per year
S5 catalyst ageing	both, separately	activity falling linearly from 0.30 to 0.10 over four years
S6 steam-to-carbon	either	steady operation at steam-to-carbon 2.5 and at 3.5

Basis of the hot-spot temperature limit. The optimisation constrains the hot-spot temperature to 1193 K (920 °C), close to the 925 °C design tube-metal temperature used by Yeh [7] for an HP-Nb tube and 46 K above the calibrated base hot spot.

Enthalpy basis of the steam credit. The steam duty charged at $\alpha < 1$ is the heat needed to raise the process steam from liquid water at 25 °C to vapour at the inlet temperature, plus the heat needed to preheat the dry feed to the inlet temperature.

Table S3: Uncertainty groups propagated in RQ4 and the distribution assigned to each input.

Group	Inputs	Distribution
kinetics	k_1–k_3, K_{CO}, $K_{\mathrm{H_2}}$, $K_{\mathrm{CH_4}}$, $K_{\mathrm{H_2O}}$	normal, from the 95 % intervals reported by Xu and Froment [1]
	effectiveness factors	log-uniform factor 0.5–2
	catalyst activity	uniform ± 0.03
heat transfer	F_{gt}, L_q, f_{htg}	multivariate normal with the joint covariance of the calibration
creep	scatter on $\log_{10} t_r$	normal, 0.181 decades, from the data-sheet band
	Larson–Miller constant C	uniform within ± 1 of 18.6
	wall thickness	uniform 12–18 mm
	tube conductivity	uniform ± 15 %
measurement	hot-spot temperature offset	uniform ± 10 K on the pyrometer reference

The flame-length variant is a bound. The structural variant $L_q\,\mathrm{load}^{0.5}$ of the uncertainty analysis is a bound, not an expectation: the length of a turbulent jet diffusion flame is nearly independent of the jet velocity for a given burner [5], so that reducing the fuel flow through all burners leaves the release profile essentially unchanged (the nominal case), whereas taking burner rows out of service at turndown, a common practice, redistributes the heat release in a way that the published data cannot constrain.

S3. Surrogate performance

The eight surrogate targets are the hot-spot temperature $T_{\mathrm{wo,max}}$ and its axial position, the outlet temperature and pressure, the dry methane slip, the net hydrogen per tube, the absorbed duty and $\log_{10} t_r$ at the hot spot. Three model families are compared with scikit-learn [6]: Gaussian-process regression with a Matérn 5/2 kernel on a 1000-point subset, histogram gradient boosting and a multilayer perceptron. Each family is fitted twice, once on the seven operating inputs alone and once with two physics-derived features added: the equilibrium methane conversion at outlet conditions and the specific duty.

Table S4: Best surrogate per target ($N_{\mathrm{train}} = 2000$ Latin-hypercube points, $N_{\mathrm{test}} = 4096$ Saltelli points): model family and feature set, test R^2 and RMSE, five-fold cross-validated RMSE, and empirical coverage (PICP) and mean width (MPIW) of the nominal 90 % CV+ conformal interval. "physics" denotes the feature set augmented with the equilibrium methane conversion and the specific duty.

Target	Unit	Model	R^2	RMSE	CV RMSE	PICP_{90} / MPIW_{90}
$T_{\mathrm{wo,max}}$	K	GP, physics	1.000	0.113	0.133	0.957 / 0.25
z/L at $T_{\mathrm{wo,max}}$	–	MLP, base	0.644	0.0319	0.0522	0.971 / 0.050
T_{out}	K	GP, physics	1.000	0.0182	0.0254	0.940 / 0.061
p_{out}	bar	GP, base	1.000	2.87×10^{-4}	5.63×10^{-4}	0.961 / 1.0×10^{-3}
CH_4 slip (dry)	%	GP, physics	1.000	1.25×10^{-3}	1.42×10^{-3}	0.953 / 4.0×10^{-3}
H_2 per tube	kmol h^{-1}	GP, base	1.000	8.47×10^{-4}	1.16×10^{-3}	0.946 / 2.9×10^{-3}
absorbed duty	W	GP, base	1.000	3.9×10^{3}	4.14×10^{3}	0.949 / 1.2×10^{4}
$\log_{10} t_r$ at hot spot	–	GP, base	0.9999	5.66×10^{-3}	5.97×10^{-3}	0.959 / 0.011

The Gaussian process dominates the other families by one to two orders of magnitude in RMSE on every continuous target, which is expected because the physics model is smooth and deterministic and the surrogate is effectively interpolating it; histogram gradient boosting is the weakest, with an RMSE of about 3 K on the hot-spot temperature. The location of the hot spot is the only target that is not learnable: it is bimodal, sitting near $z/L = 0.43$ for most inputs and moving to the outlet plateau for others, and all three families give R^2 below 0.65. It is reported for completeness and is not used in the life calculation, which requires only the maximum. Prediction takes 38 μs per point against 0.25 s for the physics model.

S4. Digitisation of the verification curves

Figure 3 of Xu and Froment [2] reports six axial profiles for one industrial reformer tube on three vertical axes: methane and CO_2 conversion on the left axis, total pressure in bar and the process-gas, inner-wall and outer-wall

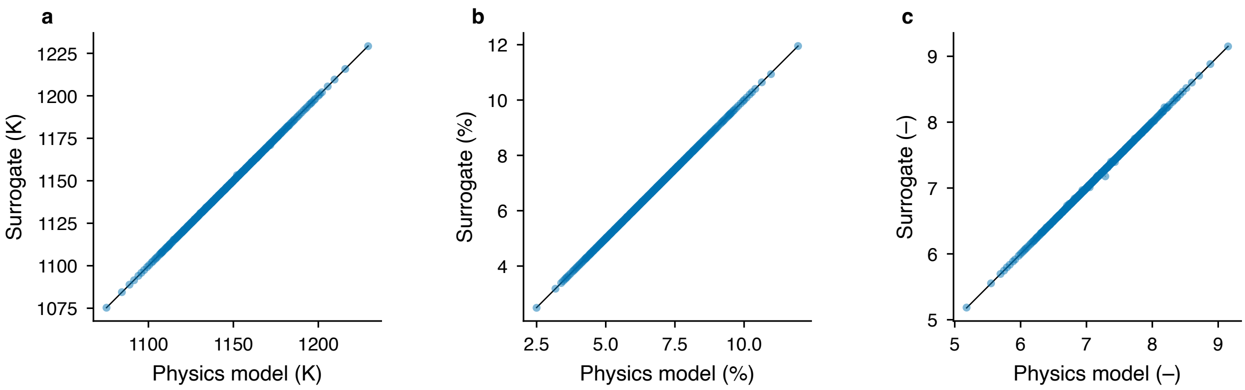

Figure S1: Parity of the selected surrogates against the physics model on the independent Saltelli test set for (a) the hot-spot temperature, (b) dry methane slip and (c) $\log_{10} t_r$ at the hot spot, with nominal 90 % CV+ conformal intervals.

Table S5: Verification of the tube model against the digitised Fig. 3 of Xu and Froment [2]: RMSE of methane conversion $x_{\mathrm{CH_4}}$, of the conversion increment $x(z) - x(0.5\ \mathrm{m})$, of the process-gas and inner-wall temperatures ($z \geq 0.3$ m) and of pressure. Bed voidage 0.526 and equivalent particle diameter 9.44 mm fitted to the pressure profile.

Configuration	RMSE $x_{\mathrm{CH_4}}$	RMSE Δx	RMSE T_g (K)	RMSE T_{wi} (K)	RMSE p (bar)
Leva–Grummer, $f_{\mathrm{htg}} = 1$, $\eta = 0.1$	0.160	–	75.4	37.5	0.04
matched $\alpha_i = 348\ \mathrm{W\,m^{-2}\,K^{-1}}$, $\eta = 0.1$	0.037	0.020	13.7	16.7	0.25
matched α_i, $\eta = 0.05/0.1$ (adopted)	0.039	0.015	14.0	16.7	0.25
matched $\alpha_i(z)$ profile, $\eta = 0.05/0.1$	0.046	0.017	17.6	16.9	0.26

Back-calculated α_i: median 348, interquartile range 313–363 $\mathrm{W\,m^{-2}\,K^{-1}}$ over 0.5–11 m.

temperatures in kelvin on two right axes. The figure was digitised with WebPlotDigitizer (version 5) from a 600 dpi scan of the printed page. Three independent axis calibrations were defined on the same image, one per vertical axis, each anchored on two gridline labels of the horizontal axis (0 and 10 m) and two labels of the corresponding vertical axis (0 and 0.8 for conversion, 24 and 29 bar for pressure, 800 and 1150 K for temperature). Points were placed manually along each curve at intervals of roughly 0.05 m of the tube length, with additional points concentrated at the discontinuity at the end of the heated length ($z = 11.12$ m), giving 118–261 points per curve.

The left axis of the original figure is labelled $x/(1+\zeta)$, where $\zeta = 0.056$ is the CO_2/CH_4 molar ratio of the feed; the digitised ordinates of the two conversion curves are therefore multiplied by $1+\zeta$ to obtain conversions referred to the equivalent methane feed. The residual uncertainty of the digitisation, estimated from the spread of repeated point placements on the gridlines, is about 2 K in temperature, 0.03 bar in pressure and 0.005 in conversion, which is small compared with the model residuals reported in the main text. The digitised curves, the axis calibrations and a tidy merged file with provenance metadata are in `data/literature_validation/digitized/` of the repository.

Two features of the digitised data are recorded but not corrected. The digitised process-gas temperature begins at 764 K whereas Table 2 of the same paper states an inlet temperature of 793 K, so all temperature comparisons in the main text exclude $z < 0.3$ m. Both conversion curves begin at about 0.057 rather than at zero; this value lies between the inlet conversion implied by the higher-alkane rule used here (0.018) and the value obtained if all C_2–C_5 carbon is converted to CO at the inlet (0.091), and the conversion increment $x(z) - x(0.5\ \mathrm{m})$ is therefore used as the primary comparison metric. The offset is consistent with a steep initial rise that the printed figure does not resolve, and it accounts for the residual of about 0.04 in absolute conversion reported in the main text.

S5. Verification and calibration detail

The pressure RMSE of the verification runs rises from 0.04 bar with the Leva–Grummer configuration to 0.25 bar with the matched film coefficient. This is not a deterioration of the friction model, which is unchanged between the two: the bed voidage and equivalent particle diameter were fitted to the pressure profile under the hotter Leva–

Table S6: Calibrated furnace and heat-transfer parameters on Plants A, B and C2 with 95 % t-intervals from the Jacobian covariance. The four-parameter fit has α_{top} at its bound with $r(L_q, \alpha_{\mathrm{top}}) = 0.985$; the adopted fit fixes $\alpha_{\mathrm{top}} = 0.182$.

Fit	Parameter	Value	95 % interval	s.e.
four parameters	F_{gt}	0.316	[0.286, 0.346]	0.014
	L_q	0.497	[0.329, 0.665]	0.081
	α_{top}	0.400	[−0.261, 1.061]	0.32
	f_{htg}	1.74	[1.427, 2.051]	0.15
	χ^2 / dof	30.5 / 23	reduced 1.32	
α_{top} fixed (adopted)	F_{gt}	0.323	[0.309, 0.338]	0.0072
	L_q	0.456	[0.434, 0.478]	0.011
	f_{htg}	1.83	[1.617, 2.052]	0.11
	χ^2 / dof	34.9 / 24	reduced 1.46	

Grummer gas, and the cooler gas of the matched run has a higher density and a lower velocity at the same mass flow.

S6. Steam-credit sweep

Table S7: Steam-credit sweep of the iso-production (H_2 = base ± 1 %) life–fuel front. $\alpha = 1$ counts firebox fuel only; $\alpha = 0$ charges steam raising and feed preheat in full. All rows are physics-verified.

α	Regime	S/C	Load	Firing	Excess air (%)	Rel. life cost	Total fuel vs base (%)	$T_{\mathrm{wo,max}}$ (K)
0.00	min life	3.99	0.96	0.909	24.6	0.059	+3.2	1103
0.00	knee	3.29	0.97	0.850	5.1	0.828	−6.6	1144
0.00	min fuel	2.66	1.00	0.888	5.0	4.539	−8.6	1171
0.25	min life	4.00	0.96	0.943	24.9	0.057	+2.3	1103
0.25	knee	3.80	0.94	0.850	5.0	0.365	−7.1	1131
0.25	min fuel	2.92	0.99	0.865	5.0	2.016	−8.9	1158
0.50	min life	4.00	0.96	1.039	25.0	0.057	+5.5	1103
0.50	knee	4.00	0.95	0.850	13.8	0.103	−7.6	1112
0.50	min fuel	3.63	0.95	0.850	5.0	0.480	−10.5	1135
0.75	min life	4.00	0.96	0.911	25.0	0.057	−6.6	1103
0.75	knee	4.00	0.96	0.850	13.5	0.105	−12.2	1112
0.75	min fuel	4.00	0.93	0.850	5.0	0.276	−14.6	1127
1.00	min life	4.00	0.96	0.958	25.0	0.057	−7.3	1103
1.00	knee	4.00	0.96	0.850	15.3	0.087	−17.7	1109
1.00	min fuel	4.00	0.93	0.850	5.0	0.276	−20.4	1127

S7. Variance decomposition

Table S8: Variance decomposition at the S1 base point: one-group-at-a-time share of the joint variance ($N = 1024$ per group) and group total-effect Sobol index computed on a Gaussian-process surrogate of the joint sample ($N = 2048$).

Group	share, $\log_{10} t_r$	S_T, $\log_{10} t_r$	share, $T_{\mathrm{wo,max}}$	S_T, $T_{\mathrm{wo,max}}$
kinetics	0.085	0.157	0.513	0.620
heat transfer	0.018	0.080	0.111	0.302
creep	0.859	0.779	0.211	0.093
measurement	0.034	0.025	0.202	0.092
sum	0.997		1.037	

The one-group-at-a-time shares sum to unity within 0.04, so the four groups act almost additively and the interaction between them is negligible at the base point. The share of the creep group in the hot-spot temperature (0.211) is

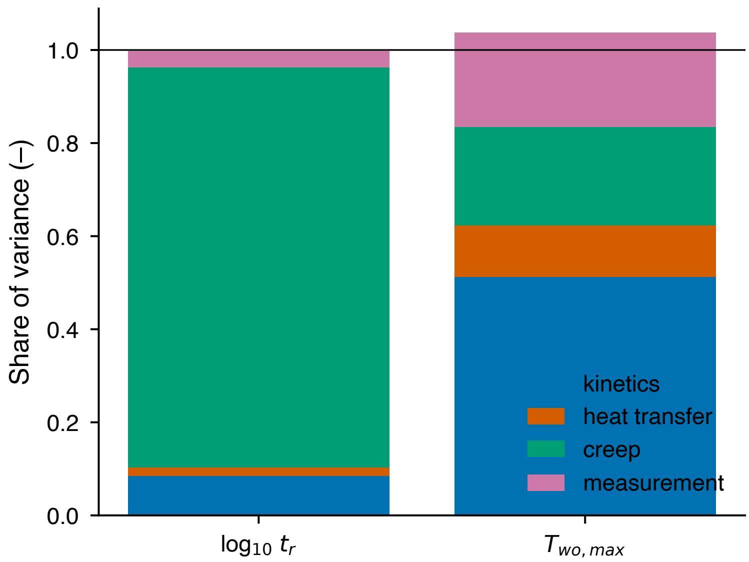

Figure S2: One-group-at-a-time shares of the variance of $\log_{10} t_r$ and of the hot-spot temperature at the S1 base point; numerical values in Table S8.

not a feedback of the creep model on the thermal solution, which does not exist, but the effect of the wall thickness, which belongs to the creep group because it also enters the hoop stress and which changes the conduction drop across the tube wall.

S8. Supplementary figures

References

[1] J. Xu, G. F. Froment, Methane steam reforming, methanation and water-gas shift: I. Intrinsic kinetics, AIChE J. 35 (1989) 88–96.

[2] J. Xu, G. F. Froment, Methane steam reforming: II. Diffusional limitations and reactor simulation, AIChE J. 35 (1989) 97–103.

[3] D. A. Latham, K. B. McAuley, B. A. Peppley, T. M. Raybold, Mathematical modeling of an industrial steam-methane reformer for on-line deployment, Fuel Process. Technol. 92 (2011) 1574–1586.

[4] M. H. Wesenberg, H. F. Svendsen, Mass and heat transfer limitations in a heterogeneous model of a gas-heated steam reformer, Ind. Eng. Chem. Res. 46 (2007) 667–676.

[5] W. R. Hawthorne, D. S. Weddell, H. C. Hottel, Mixing and combustion in turbulent gas jets, Third Symposium on Combustion and Flame and Explosion Phenomena 3 (1949) 266–288.

[6] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, É. Duchesnay, Scikit-learn: machine learning in Python, J. Mach. Learn. Res. 12 (2011) 2825–2830.

[7] C.-L. Yeh, Numerical analysis of the combustion and transport phenomena in a reformer tube, Appl. Sci. 11 (2021) 231.

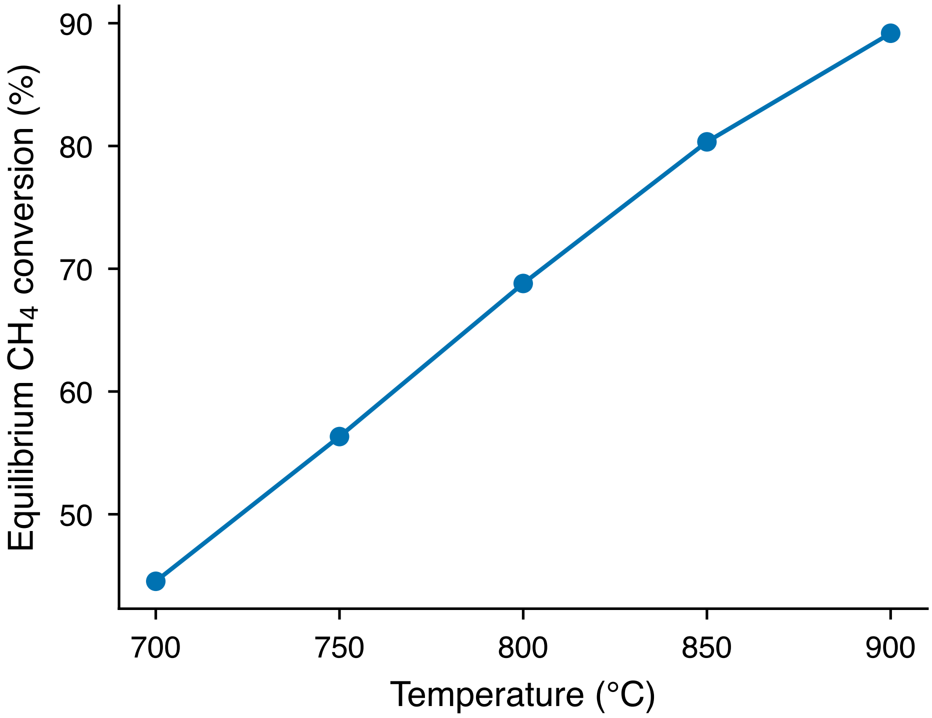

Figure S3: Equilibrium methane conversion of a CH_4/H_2O mixture at 25 bar and a steam-to-carbon ratio of 3, computed with Cantera and the GRI-Mech 3.0 thermodynamic data. This is the thermodynamic ceiling against which the reactor model is checked; at 850 °C the equilibrium conversion is 0.80 and the dry hydrogen fraction 73 %.

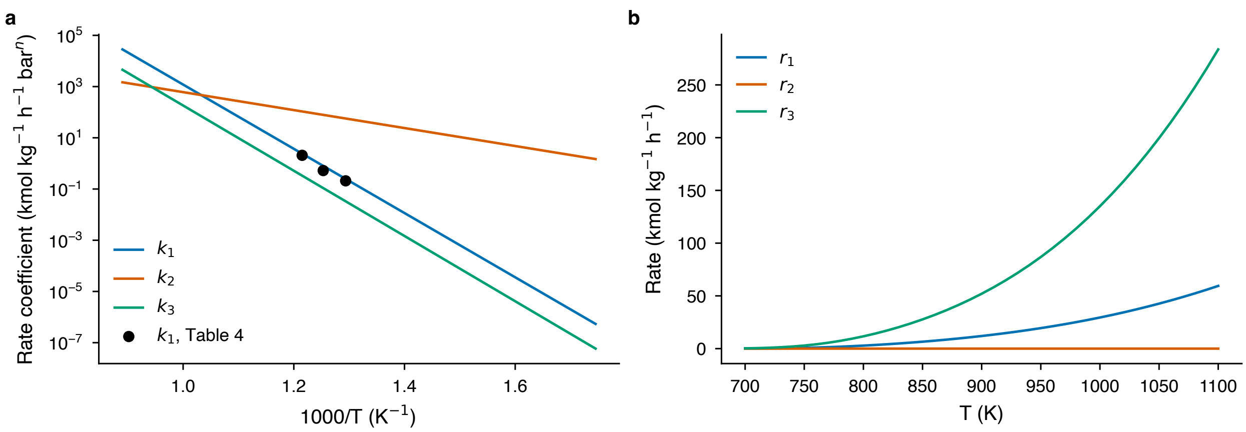

Figure S4: Xu–Froment kinetics as implemented: (a) Arrhenius plot of the rate coefficients k_1, k_2 and k_3 against $1000/T$, with the per-temperature estimates of Table 4 of [1] overlaid, and (b) the three reaction rates against temperature at 25 bar for a CH_4:H_2O:H_2 feed of 1:3:0.1.

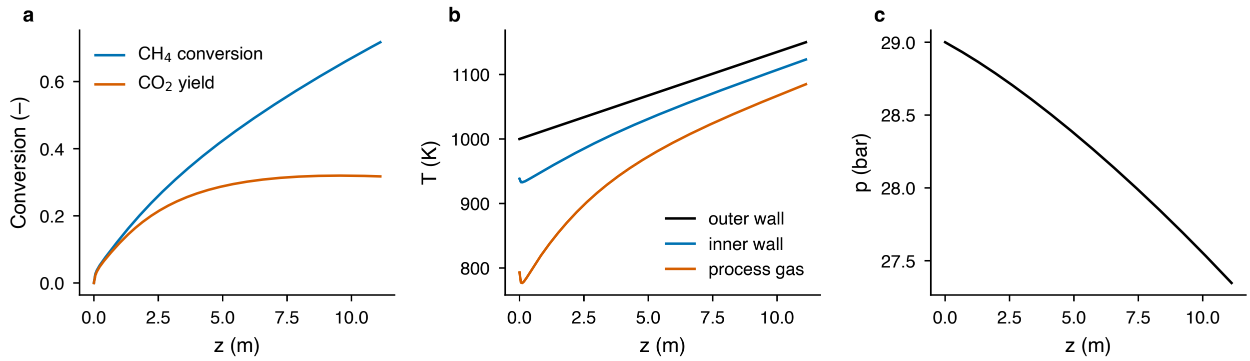

Figure S5: Stand-alone tube model with a prescribed linear outer-wall profile: (a) methane conversion and CO_2 yield, (b) process-gas, inner-wall and outer-wall temperatures and (c) pressure along the tube for the feed of Table 2 of [2]. This run precedes the verification of the main text and uses no fitted parameters.

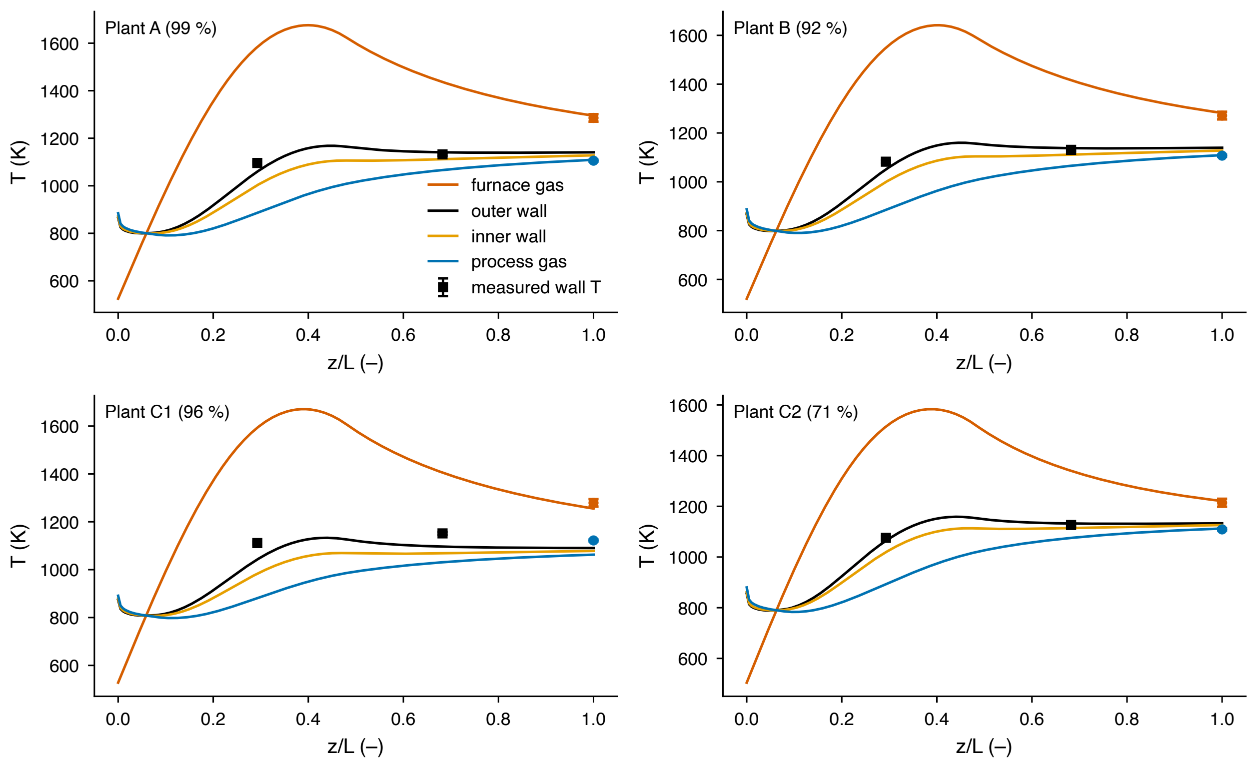

Figure S6: Coupled furnace–tube model before calibration, with the parameter values of Table 6 of [3] transferred directly: axial temperature profiles for the plant cases and the measured wall temperatures. Three of the four cases are reproduced within the measurement uncertainty without any fitting.

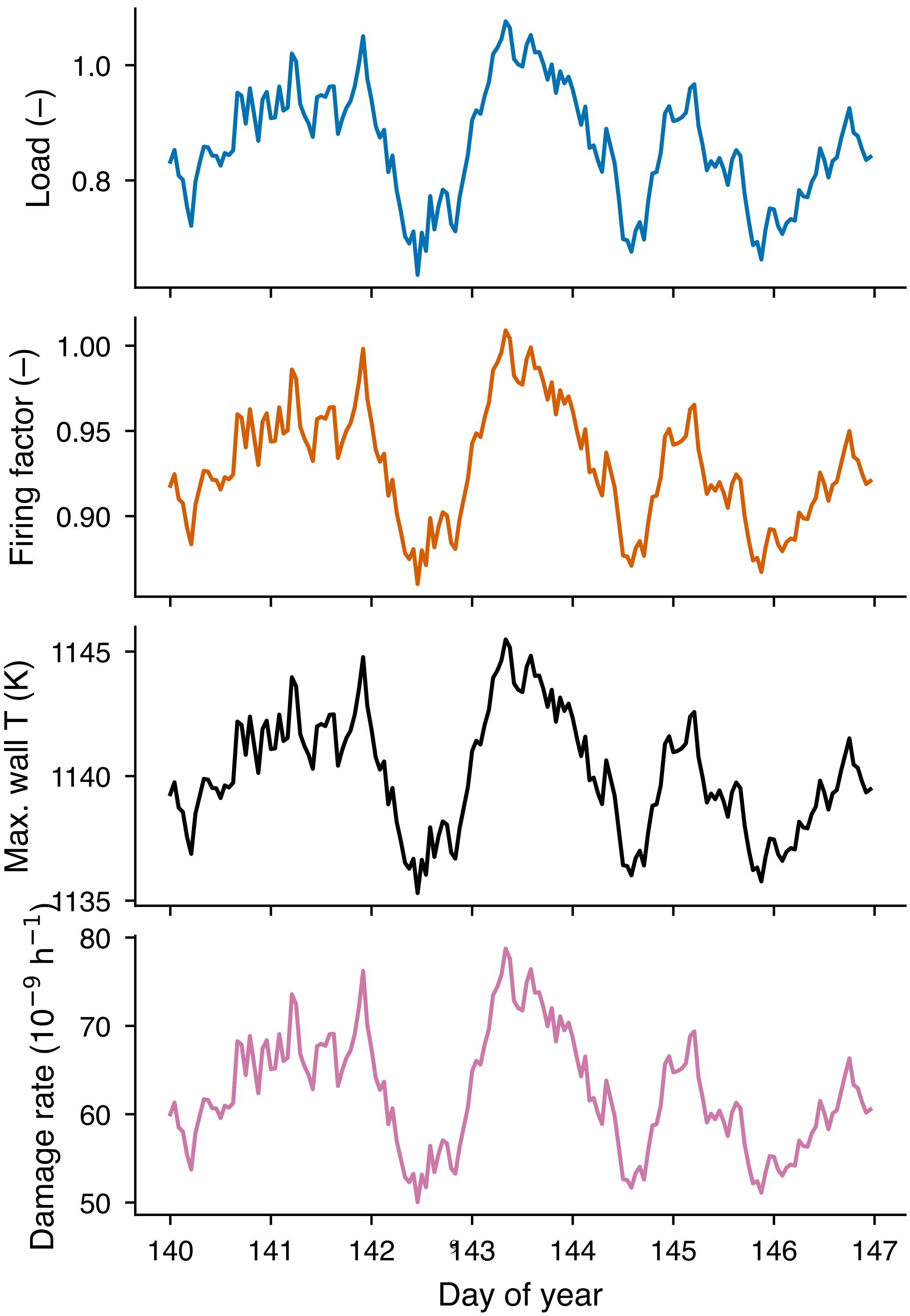

Figure S7: One representative week of the renewable-following scenario S3: load, firing factor, hot-spot temperature and hourly damage rate. The damage rate follows the hot spot through the Larson–Miller relation and is therefore strongly non-linear in the load.